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The National Parks System's Effect on Local Economies for Rural and Urban Counties

Introduction

This project seeks to explore the effects on which proximity to a national park can have on local economies, with a focus on an urban-rural divide. National parks present significant tourism opportunities across the country which bring significant economic activity to the local gateway communities. However, this effect is unlikely to be consistent across the country as some areas may be more dependent on tourism if there is less ability for other industries to take place there. Urban areas often have more diverse economic sectors than rural areas as many rural areas do not have the same level of capacity for industries. This becomes interesting when applying to rural areas near national parks which may see a lot of their industry from tourism to the park since the park dominates the local landscape, precluding moves toward diversifying the local economy. It is possible there is a level of spatial dependence present within the relationship of national parks and their surrounding communities, with some level of the economy presumably depending on tourism to the park. The main motivation between this project is to assess the level on which national parks can affect a local economy which has become especially important in recent years as park budgets tend to be more up in the air, which may also come with significant effects on gateway economies. For example, Indiana Dunes National Park is located relatively close to the Chicagoland area, and while the park certainly brings in tourists, its urban nature seems like it would be less important to have a park there for the locals.

Alternatively a park like Grand Canyon National Park, which exists in a more remote area, sees many surrounding communities have the infrastructure to aid in supporting the park's many tourists, which may suggest a greater level of economic dependence. Essentially, in accounting for how dependent more rural areas can be on tourism to the parks, we can analyze the potential economic importance which the parks, and their budgets, bring to local economies.

The central hypothesis of this project is that counties near national parks will be more economically dependent on tourism if they are located in a rural area as opposed to urban areas. It is expected that rural areas will exhibit higher concentration of tourism related activity and are thus disproportionately more dependent on the tourism industry than other local sources of business and employment. In contrast, urban areas will likely see more diversified levels of economy in which the national park makes up a considerably smaller percentage of activity. This spatial analysis seeks to examine whether local economies will share similar spatial dependence characteristics among gateway communities as well as spatial heterogeneity for different park regions across the country. Tourism would be expected to be spatially clustered around national parks and high tourism counties are likely close to other high tourism counties. Additionally, the effect of tourism dependence is expected to demonstrate high spatial heterogeneity as the effect is likely different across rural and urban counties, with rural counties exhibiting a stronger response.

Several potential sources of bias and variation are worth consideration for interpreting the results of the project. Notably, there is a vast variation of the relevance for national parks across the country, meaning that not all parks are created equal. Some parks are simply more popular than others and this project does not consider popularity. Additionally, there are many remote national parks which simply do not have the land for tourism infrastructure even if they are in a

rural area. Specifically, Alaskan National Parks come to mind as there are millions of acres of park land in the state that does not have any level of tourism due to their remote nature even though it is still a national park. Variation across different parks includes different levels of popularity and accessibility. Additionally, there are some limitations with the data to be acknowledged. The method in which this project measures tourism dependence through NAICS codes uses more general classification and may not actually capture the effect of the park or may omit indirect effects of tourism as well. This can cause the relationship to the tourism industry to be different than expected due to the presence of other attractions as well. This analysis also does not take into consideration the historical economic activities that an area may have which could change the effect independently of the park.

Methods

Data

Data was taken from multiple sources to get park and county boundaries, local economic sector data, and rural/urban classifications. National Park Boundaries were taken directly from the NPS GIS Portal in the NPS Boundary Data Shapefile (2025). The dataset had polygon data on all the national parks as well as other national areas and monuments which could then be mapped in space. County boundary data was from the US Census Bureau's 2025 TIGER/Line Shapefiles: Counties (and equivalent), and was taken directly from the census's website. This dataset contained polygon data for all US counties which could then be used with the national park boundaries. Economic data was taken from the Bureau of Economic Analysis which gives industry information in the

NAICS code composition of local economies, which was also taken directly from the BEA website. It should also be noted that the county level NAICS code data was discontinued in 2023, which is the last available data for use in this project. Lastly, a classification system from the USDA was used to determine if counties were rural or urban for use in this project. The Rural-Urban Continuum codes were applied to this project and taken directly from the USDA website.

This data has been further simplified for use in this project as well. The National Park boundary data was simplified to only include national parks as well as omitting parks from US territories and Hawaii. Hawaii was excluded due to the state's island nature and small area which would likely interfere with seeing the effect of national parks on surrounding areas. Additionally, the high tourism relevance to Hawaii beyond the national parks would not give a good insight into the actual effect of the national parks, thus the decision to omit Hawaii's two national parks was made. Similarly, the National Park of American Samoa and Virgin Islands National Park were not included in this project. A threshold of 30 miles for areas around the national park was selected to define gateway counties as was designated as a binary variable in the data. Tourism data was used to compute the tourism share variable which is defined as the percentage of employment that belongs to tourism related NAICS codes compared to total employment. Lastly, a rural or urban designation was given to counties based on the RUCC code with codes 1-3 representing metro areas and 4-9 representing non-metro areas. Counties were assigned with a binary variable for rural and urban designation based on the metro or non-metro value from the RUCC.

Descriptive Statistics

A summary table was generated to show the distribution of tourism share among gateway counties, accounting for a Rural and Urban, as well as a histogram to aid in visualizing the distribution. The table shows the mean and standard deviation of tourism dependence, as well as each quartile point, which can be visualized through the histogram as well.

Spatial Analysis

Various tools of spatial analysis were used throughout this project to test the hypothesis. Choropleth maps were used to visualize the spatial distribution of urban/rural divides as well as tourism dependence. The first map demonstrates the rural/urban divide, as defined by the RUCC codes, among gateway counties as it is seen across the US, exhibiting the locations of areas of the National Park System that can be classified as rural. The second map is based on the percentage of tourism found in the county-level economies and is displayed as a standard deviation with intervals in terms of standard deviations away from the mean for all gateway counties. This style of map was chosen to combat the heavy outliers which skew intervals away from the lower percentages found around the mean. Together, these maps give a benchmark to the national distribution of rural/urban gateway counties and their tourism levels.

Global Moran's I was used to test for spatial autocorrelation of tourism dependence among gateway counties to national parks. Given the distance which exists between the national parks, the spatial weights matrix is produced as multiple

disconnected parts. This means the Global Moran's I value does not account for the entire population of counties but instead only relates the regional groupings of gateway counties. Given the result from this test, LISAs will be used to describe the types of clustering found surrounding national parks. Together these tests assess the degree of spatial dependence in tourism reliance among national park gateway counties.

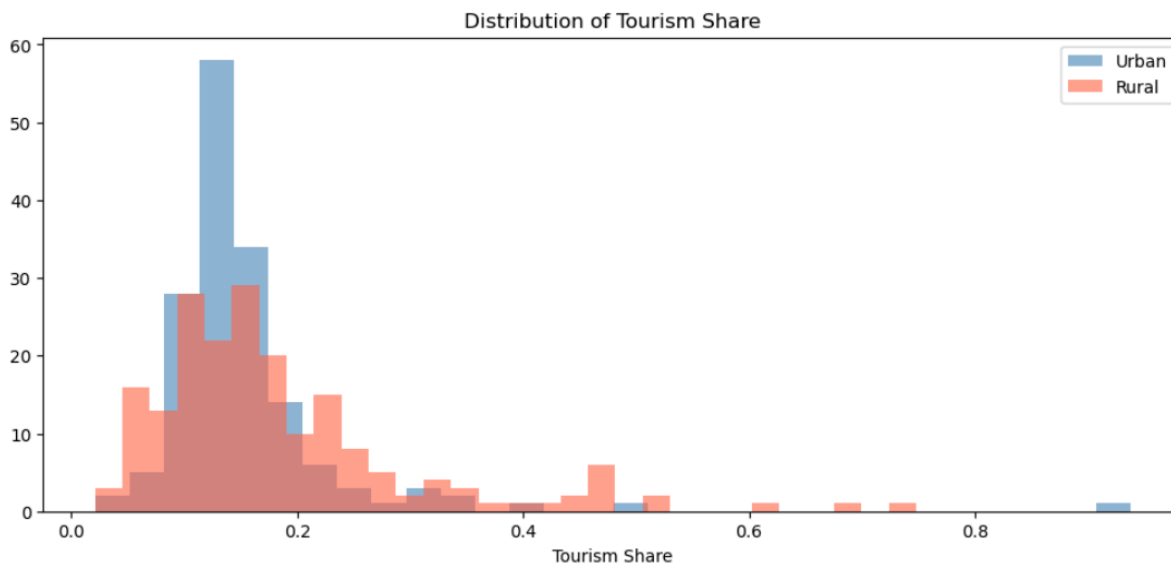
Lastly, a spatial regression as used to formally test for rural gateway counties exhibit higher dependence on tourism than urban gateway counties. An OLS baseline regression was performed then subsequent diagnostics testing to account for the spatial dependence of the data.

Results

The results from each part of this project are as follows:

Descriptive Statistics

The descriptive statistics for the relevant data give the following table and histogram.

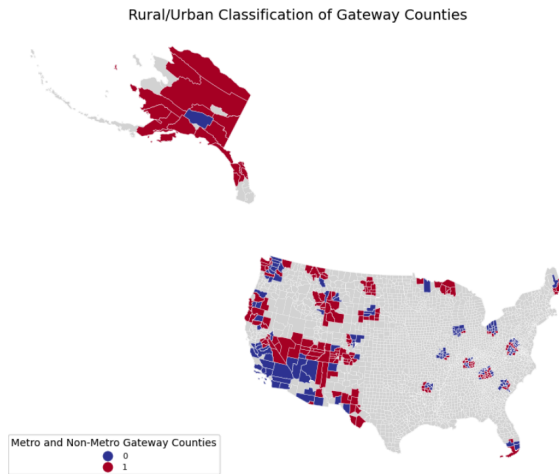


	count	mean	std	min	25%	50%	75%	max
Rural								
0	159.0	0.153999	0.086738	0.021277	0.120706	0.139531	0.161133	0.937957
1	194.0	0.183205	0.119149	0.020951	0.107969	0.156812	0.217642	0.747871

The histogram demonstrates a right skewed distribution for both urban and rural gateway counties for their tourism share. The mean tourism share for rural is slightly higher than urban and also has a higher standard deviation. This demonstrates that rural economies have a larger spread of tourism dependence as well. Overall, both of these indicate there may be a notable difference in tourism dependence among rural and urban gateway counties.

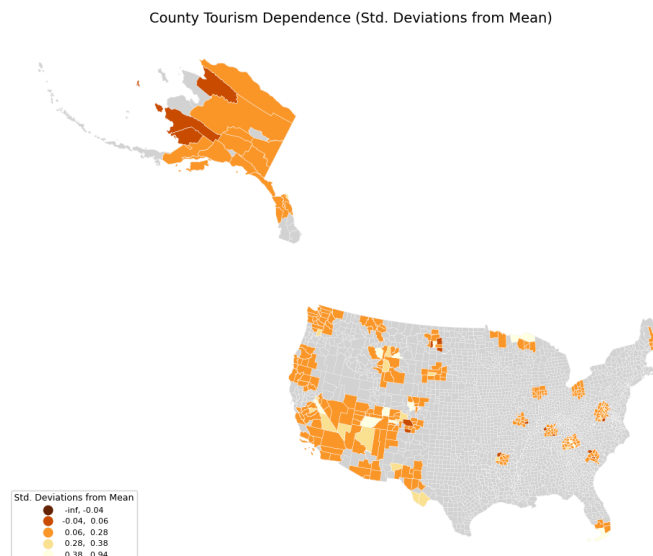
Choropleth Maps

Map 1: Classification of Gateway Counties



The first map demonstrates what each gateway county is designated as, metro and non-metro. This indicates that parks exist in both rural and urban settings across different regions of the country, reflecting some validity into testing the hypothesis. Additionally, I think it is interesting that there are not many areas in which the park is surrounded by only rural or urban counties.

Map 2: Standard Deviations from Mean Tourism Dependence



The second map demonstrates the spread of tourism dependence across gateway counties. The lighter colors indicate very high tourism dependence and the darker colors are the opposite. This suggests that most of the counties appear to be around the mean, but there are several higher values present throughout the country as well. Purely based on looking at the map, many more empty areas of the country seem to have lighter colored counties.

Global Moran's I

The Global Moran's I value was 0.2189, indicating moderate positive spatial correlation with strong statistical significance, $p = 0.0010$. This indicates that there is an element of spatial dependence of tourism share across gateway counties and that it is not random. The cluster of high tourism dependent gateway counties with other high tourism dependent gateway counties is important and significant now to this project. This finding can be used as information and justification for local statistics and spatial regression.

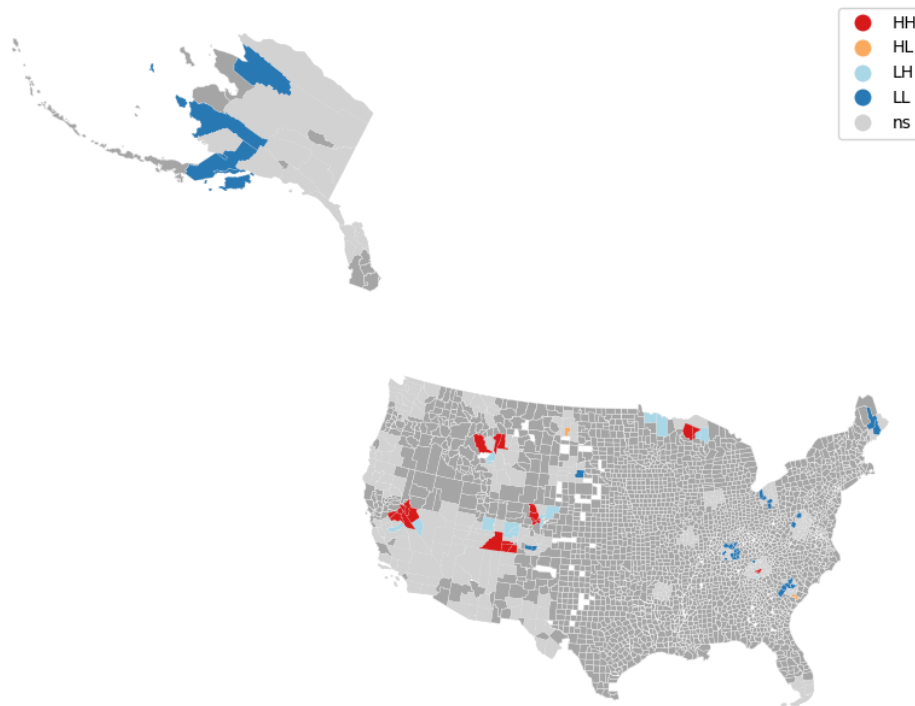
LISAs

The clusters found as a result of testing for LISA were:

NS	722
LL	35
HH	22
LH	17
HL	2

There were several significant examples of clusters for gateway counties for almost every category, the HL clusters being the only one with less than 17. The HH clusters represent counties in areas of high tourism dependence as well. Interestingly, when looking at rural counties, they make up 81.8% of the HH clusters, despite making up 54% of the all gateway. This indicates that HH clusters are disproportionately rural counties and demonstrates that rural

counties typically compose greater areas of large tourism dependence.



The map above depicts the locations of significant clusters across the country. Many of the LL clusters appear to be in the Eastern US while the HH clusters appear to be more western. This could reflect the proximity of Eastern US parks to urban areas compared to the more rural parks of the west.

Spatial Regression

The preliminary OLS regression results yield a coefficient of 0.0292 for rural gateway communities to explain tourism dependence with an r-squared of 0.018. While these results are weak, the test diagnostics indicated there was spatial dependence among residuals resulting in the use of a spatial error model for regression. The Spatial Error Model yielded a coefficient of 0.0226 but was not statistically significant at $p = 0.056 > 0.05$. Additionally, the lambda

coefficient from the regression came back as very statistically significant, indicating that there are unmeasured factors which explain the difference beyond rural classification. The spatial regimes model estimated a mean tourism share of 18.3% for non-metro and 15.4% for metro counties. The Chow test confirmed that these regimes are statistically distinct at $\chi^2 = 7.079$ and $p = 0.008$), providing some formal evidence of spatial heterogeneity in tourism dependence across rural and urban gateway communities.

Conclusion

Overall, there was not strong evidence to suggest that my hypothesis of rural gateway counties being more dependent on tourism was correct. The Global Moran's I did not suggest there was strong spatial autocorrelation, but instead the connection was moderate at best. The LISAs did indicate there was some level of merit to the argument given the prevalence of rural counties among HH clusters. While this would show that rural areas surrounding parks have some level of dependence on tourism, the rest of the cluster failed to demonstrate anything significant as well, meaning the HH clusters situation is not likely to have meaningful consequences. The LISA map gives some interesting insight into where the notable clusters exist and I think it is interesting there is largely an east-west divide visually, which is interesting considering the east is generally more urban, but this still likely fails to have meaningful consequences. The spatial regression also yielded a largely insignificant result as the coefficient for rural gateway counties on tourism dependence was not statistically significant, and even then only would have accounted for a 2% difference. The spatial regime test did show that there was a significant difference between urban and rural gateway counties however, which may indicate there are further explanations to be explored on the differences between the groups and overall does solidify that the rural gateway counties have a higher dependence on tourism. Overall, this analysis fails to suggest that rural

national park gateway counties have a larger dependence on tourism than urban gateway counties.

One interesting spatial pattern which came out of this project was that there was not a strong Moran's I among gateway counties. I would have expected this number to be very high, especially considering they were only weighted against other nearby gateway counties, but it seems that quirk had a different effect. Additionally, I expected that national park gateway counties would be more dependent on tourism as a whole than they actually were. It appears that national parks may not bring as much economic activity to local parks as I expected, though it was still larger than average. The choropleth maps also surprised me as I did not expect gateway counties to seemingly vary in the same park as I did not expect to see so many rural and urban counties next to each other, presumably around the same park. This may indicate there are better ways to classify counties as rural or that parks do not often exist in purely rural or urban spaces. I think the LISA clusters also were interesting as there were far more low values than I expected, especially that there were a lot of LH. I think it would be interesting that some gateway counties would not be very dependent on tourism despite surrounding areas that are. This might indicate that even though they are near national parks, certain counties are better equipped for tourism and even access to the park. Overall, despite the large insignificance of the result, there were many interesting spatial patterns which occurred and may warrant further investigation.

Given the results, I think my next step would be to begin reworking the project to cover some of the holes I faced. For example, there may be a better way to define rural counties in the context of national parks, considering how mixed some of the areas were. Furthermore, there is probably a more elegant definition to determining gateway counties than a 30 mile buffer, perhaps looking at what counties actually hold park entrances. The definition for tourism

dependence could probably use some work as well since the NAICS codes probably do not accurately reflect the economic activity generated from the park. If I had more time and resources, I might work on establishing a way to find the economic effect of national parks. There may be other topics worth considering as well if this idea does not end up going anywhere which showed promise from working on this project. For example, it may be interesting to analyze park gateways when compared to other areas, not just other gateways. I think after refining the definitions and data used for some of my variables it would probably be worth going forward to public groups for further engagement, regardless if the results were what I wanted. Going forward with insignificant results can still be important to the overall goals of research groups and it is possible my work could get other people thinking about national parks as well. Overall, the results of my project could prompt reworking my approach and seeking other input to eventually get to a strong research project.

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